# Efficacy and Safety of Rituximab in the Management of Pediatric Systemic Lupus Erythematosus: A Systematic Review

Ines Mahmoud, PhD[1], Manel Jellouli, MD[2], Imen Boukhris, PhD[3], Rim Charfi, MD[4], Aicha Ben Tekaya, MD[1], Olfa Saidane, MD[1], Maryem Ferjani, MD[2], Yousra Hammi, MD[2], Sameh Trabelsi, PhD[4], Narjess Khalfallah, PhD[3], Rawdha Tekaya, PhD[1], Tahar Gargah, PhD[2], and Leila Abdelmoula, PhD[1]

**Objectives** To evaluate the efficacy and safety of rituximab for treating pediatric systemic lupus erythematosus (pSLE).

**Study design** We performed a systematic review to evaluate the efficacy and safety of rituximab in children with pSLE. Data from studies performed before July 2016 were collected from MEDLINE, the Cochrane Library, Scopus, and the International Rheumatic Disease Abstracts, with no language restrictions. Study eligibility criteria included clinical trials and observational studies with a minimal sample size of 5 patients, regarding treatment with rituximab in patients with refractory pSLE (aged <18 years at the time of diagnosis). Independent extraction of articles was performed by 2 investigators using predefined data fields.

**Results** Twelve case series met the criteria for data extraction for the systematic review with a good quality assessment according to an 18-criteria checklist using a modified Delphi method. Among them, 3 studies were multicenter and 3 were prospective. The total number of patients was 272. Studies collected patients with active disease refractory to steroids and immunosuppressant drugs. Refractory lupus nephritis was the most common indication (33%). Acceptable evidence suggested improvements in renal, neuropsychiatric and haematological manifestations, disease activity, complement and anti-double stranded Desoxy-Nucleo-Adenosine, with a steroid-sparing effect. However, there was poor evidence suggesting efficacy on arthralgia, photosensitivity, and mucocutaneous manifestations of SLE in children. An overall acceptable safety profile with few major adverse events was shown.

**Conclusion** Rituximab exhibited a satisfactory profile regarding efficacy and safety indicating that this agent is a promising therapy for pSLE and should be further investigated. *(J Pediatr 2017;187:213-9).*

The incidence of pediatric systemic lupus erythematosus (pSLE) is 0.3-0.9/100 000 children per year, with a prevalence of 3.3-8.8/100 000 children.[1] pSLE is associated with more severe and active disease compared with systemic lupus erythematosus (SLE) in adults. In particular, there is a higher incidence of renal and central nervous system involvement.[2,3] Immunosuppressive therapies are widely used but fail to show efficacy in severe forms of the disease.

B-lymphocyte function is a major component of SLE pathogenesis.[4,5] Rituximab (RTX) is a chimeric monoclonal IgG1 kappa antibody that binds specifically to the CD20 antigen and mediates B-cell depletion, thereby preventing the renewal of autoantibodies and antigen presentation by pathogenic B cells.[6,7] Recent reports have suggested the use of RTX for patients with SLE who are refractory to standard treatment approaches; however, RTX therapy has not been approved for this indication.[8] To understand and interpret the available evidence, we conducted a systematic review to evaluate the efficacy and safety of RTX therapy in children with pSLE.

## Methods

This systematic review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. The search strategy was performed in the digital databases of MEDLINE, Scopus, Web of Knowledge, and the Cochrane Library. We also searched the electronic databases of the American Society of Rheumatology, the European League Against Rheumatic Disease, and the French Society of Rheumatology. We performed a hand search by reviewing the references of the included studies. We searched all databases from their earliest records up to July 2016. Keywords included terms and synonyms for RTX, lupus, and child.

| | |
|---|---|
| LUNAR | Lupus Nephritis Assessment with Rituximab |
| pSLE | Pediatric systemic lupus erythematosus |
| RTX | Rituximab |
| SLE | Systemic lupus erythematosus |

From the [1]Department of Rheumatology; [2]Department of Pediatric Nephrology; [3]Department of Internal Medicine, Charles Nicolle Hospital, Faculty of Medicine, Manar University, Tunis, Tunisia; and [4]Department of Clinical Pharmacology, National Centre of Pharmacovigilance, Faculty of Medicine, Manar University, Tunis, Tunisia

The authors declare no conflicts of interest.

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http://dx.doi.org10.1016/j.jpeds.2017.05.002

### Selection Criteria

We included both clinical trials and observational studies of pediatric patients (aged <18 years) with SLE treated with RTX, published in full text or abstract. There was no language restriction. A minimum sample size of 5 patients was required for inclusion. Studies including both children and adults were excluded if the data for children could not be extracted separately.

### Data Extraction

Two investigators reviewed titles and abstracts and then selected potential manuscripts for retrieval. The same investigators collected the data from the potential studies by applying the aforementioned selection criteria. A standardized data collection form was used to extract the following information: type of study and use of controls, number of patients, patient age, systemic involvement of SLE, dose and administration schedule of RTX, other medications, and outcome measures assessing the efficacy and/or safety and duration of response. The definition of efficacy for nonrenal manifestations included activity index scores, SLE flares, steroid-sparing effects, and effects on laboratory measures. Of the several definitions of primary outcome used, the most common was an improvement in clinical disease activity according to the SLE Disease Activity Index. Definitions of partial and complete remission used in the various studies are listed in **Table I** (available at www.jpeds.com). Safety records included reports of infections and malignancies.

### Risk of Bias Assessment

We used an 18-criteria checklist developed by a panel of 7 Health Technology Assessment researchers using a modified Delphi method to perform a quality assessment on case series studies.[21] Checklist items included study objective, study population, interventions and cointerventions, outcome measures, and statistical analysis. Two investigators (I.M. and M.J.) rated each study working independently; any disagreements were resolved by consensus. The criterion for inclusion of a study was a score of ≥14 out of a possible 18.

### Pooling of Data

Given the differences in study design, interventions, and reported outcome measures, a meta-analysis was deemed inappropriate, and we focused on describing the studies in evidence tables with their results and a qualitative synthesis.

## Results

Our search of the databases yielded 549 citations, of which 12 studies[9-20] (with a total of 272 patients) met the inclusion criteria, including 10 complete articles and 2 abstracts (with 32 patients).[10,18] The **Figure** (available at www.jpeds.com) illustrates the literature selection process. All of the studies were case series and did not include a control group.

### Quality Assessment

All of the studies clearly described an objective, patient inclusion/exclusion criteria, patient characteristics, and main findings (**Table II**). Likewise, all studies had clearly defined outcomes. Three studies were multicenter,[14,15,20] 3 studies collected patient data prospectively,[10,13,18] and 3 studies involved consecutive enrollment of patients.[11,13,18]

### Epidemiologic Characteristics

British, American, and Canadian studies were predominant (a total of 162 patients; 60%).[9,13,15,17,19] The patients were mostly female (80%) and ranged in age from 6 to 28 years. The mean age at onset of SLE was reported in 6 studies and ranged from 7.8 to 12.5 years. The mean duration of disease was reported in 9 studies and ranged from 1.4 to 4.7 years. Regarding previous therapies, all of the patients had failed to respond to cyclophosphamide and mycophenolate mofetil, and most had failed to respond to at least 1 other immunosuppressive agent (ie, azathioprine, intravenous immunoglobulin, methotrexate, cyclosporine, tacrolimus, or thalidomide). The duration of follow-up ranged from 1 month to 36 months. Study characteristics are summarized in **Table III**.

**Table II. Quality assessment of the studies according to the modified Delphi method**

| Studies | Objective clearly described | Inclusion/exclusion criteria clearly described | Multicenter study | Data collected prospectively | Patient characteristics clearly described | Outcomes clearly defined | Main findings clearly described |
|---|---|---|---|---|---|---|---|
| Willems et al (2006)[20] | Y | Y | Y | N | Y | Y | Y |
| Nwobi et al (2008)[17] | Y | Y | N | N | Y | Y | Y |
| Podolskaya et al (2008)[19] | Y | Y | N | N | Y | Y | Y |
| Jansson et al (2011)[14] | Y | Y | Y | N | Y | Y | Y |
| Su et al (2012)[10]* | Y | Y | NR | Y | Y | Y | Y |
| Pavon-Sanchez and Sánchez-Sánchez (2013)[16] | Y | Y | N | N | Y | Y | Y |
| Alexeeva et al (2013)[18]* | Y | Y | N | Y | Y | Y | Y |
| Lehman et al (2014)[13] | Y | Y | N | Y | Y | Y | Y |
| Ale'ed et al (2014)[12] | Y | Y | N | N | Y | Y | Y |
| Olfat et al (2015)[11] | Y | Y | N | N | Y | Y | Y |
| Tambralli et al (2015)[9] | Y | Y | N | N | Y | Y | Y |
| Watson et al (2015)[15] | Y | Y | Y | N | Y | Y | Y |

*N*, no; *NR*, not reported; *Y*, yes.
*Abstract.

**Table III. Characteristics of studies**

| Studies | Country | No. of patients | M/F, n | Age at SLE onset, y, mean ± SD | Disease duration, y, range, mean ± SD, or median (range) | Age of initiation of RTX, y, range, mean ± SD, or median (range) | Previous treatments | Concomitant treatments | Indication for RTX |
|---|---|---|---|---|---|---|---|---|---|
| Willems et al (2006)[20] | France | 11 | 0/11 | | 1-9 | 8.3-19 | CS, CYC, AZA, MMF, IVIG, MTX, HCQ, PE | CYC, AZA, MMF, IVIG, MTX, HCQ | Nephritis, PHT, pericarditis, hemolytic anemia, thrombocytopenia |
| Nwobi et al (2008)[17] | US | 18 | 2/16 | 10.7 ± 2.5 | 3.1 ± 2.5 | 14.2 ± 3.3 | CS, CYC, AZA, MMF, CSA, IGIV | HCQ, CS | Nephritis, vasculitis, arthritis, ataxia, thrombocytopenia, pht, cytotoxicity, serositis ophthalmic, vasculitis, hemolytic anemia |
| Podolskaya et al (2008)[19] | UK | 19 | 2/17 | NR | 4.06 (0.1-7.9) | 14 (6-16) | CS, CYC, AZA, MMF, IVIG , MTX, HCQ | CS, CYC, AZA, MMF, HCQ | Generalized symptoms, hematologic, renal, joint, skin, cerebral disease |
| Jansson et al (2011)[14] | Germany | 19 | 4/15 | 11.3 | - | 15.3 (8.3-18.7) | NR | NR | NR |
| Su et al (2012)[10]* | China | 20 | 7/13 | NR | 3.1 ± 2.5 | NR | NR | NR | NR |
| Pavon-Sanchez and Sánchez-Sánchez (2013)[16] | Mexico | 8 | 2/6 | 11.3 ± 1.03 | 1.7 ± 0.5 | | CS, CYC, AZA, MMF, MTX, IVIG | NR | NR |
| Alexeeva et al (2013)[18]* | Russia | 12 | 1/11 | NR | 3.6 (0.3-13.2) | 14.6 (10-17.5) | CYC, MMF, IVIG, CS, PE, | CS, immunosppressive agents | Nephritis |
| Lehman et al (2014)[13] | US | 12 | 3/9 | 12.5 | NR | 16 (10-28) | NR | CYC | Nephritis |
| Ale'ed et al (2014)[12] | Saudi Arabia | 16 | 3/13 | 7.8 ± 3.3 | 4.7 ± 3.2 | NR | CS, AZA, CYC, MMF, CSA, MTX | AZA, CYC, MMF, CSA, MTX | Nephritis, cerebritis, polyarthritis, mucocutaneous, thrombocytopenia |
| Olfat et al (2015)[11] | Canada | 24 | 6/18 | NR | NR | 13.2 (10.5-15.9) | CS, IVIG, AZA, CYC, MMF, MTX, splenectomy | CS, MMF, CYC | Thrombocytemia, hemolytic anemia, thrombocytopenia, hemolytic anemia |
| Tambralli et al (2015)[9] | US | 50 | 9/41 | NR | 1.6 ± 1.3 | 13.6 ± 3.5 | CS, AZA, HCQ, MTX, MMF, biologicals, CYC | AZA, HCQ, MTX, MMF, biologics, CYC | NR |
| Watson et al (2015)[15] | UK | 63 | 13/50 | 12.2 (9-13.9) | 1.4 (0.2-3) | 14.4 (12-15.5) | CS, CYC, AZA, MMF, CSA, MTX, PE, tacrolimus, infliximab, etanercept, thalidomide | CYC, IVIG, thalidomide, biologicals, PE, CS | Nephritis, vasculitis, cerebral disease |

*AZA*, azathioprine; *CS*, corticosteroids; *CSA*, cyclosporine; *CYC*, cyclophosphamide; *F*, female; *HCQ*, hydrochloroquine; *IVIG*, intravenous immunoglobulin; *M*, male; *MMF*, mycophenolate mophetil; *MTX*, methotrexate; *NR*, not reported; *PE*, plasma exchange; *PHT*, pulmonary hypertension; *RTX*, rituximab; *SLE*, systemic lupus erythematosus.
*Abstract.

### RTX Administration and Disease Indication

Refractory lupus nephritis was the most common indication for RTX therapy (7 studies). Otherwise, children were treated for various disease manifestations, including cytopenia (5 studies), vasculitis (3 studies), arthritis (3 studies), serositis (2 studies), neuropsychiatric disease (4 studies), skin disorders (2 studies), pulmonary hypertension (2 studies), ophtalmic disorders (1 study), and general symptoms (1 study). RTX dosage data were reported in 11 studies (264 patients). There was wide heterogeneity in reported RTX regimens. The most common regimen was RTX infusion at a dose of 375 mg/m$^2$/week.[10-12,14,17,18,20] The number of doses per course ranged from 2 to 4, and the number of courses ranged from 1 to 12. Four studies reported a RTX dose of 750 mg/m$^2$ (maximum 1 g) given as 2 intravenous infusions separated by approximately 14 days.[9,13,15,19] Watson et al[15] reported the highest number of courses (n = 104). Regarding concomitant therapies, 7 studies[9,11-13,15,19,20] reported patients with severe disease receiving RTX combined with intravenous cyclophosphamide. In most studies, glucocorticoids and nonbiological immunosuppressive therapy were associated. **Table III** presents indications for RTX and concomitant treatments.

### Global Disease Activity

Three studies reported short-term (1-8 months) statistically significant improvements in disease activity (50%-96% complete remission and 36%-56% partial remission).[10,14,15,17] Long-term (≥12 months) outcomes were reported by Podolskaya et al[19] (52% complete remission and 43% partial remission). Six of the 12 studies assessed disease activity using the SLE Disease Activity Index and described statistically significant improvement at 1, 3, 6, and 12 months[10,12,13,16-18] (**Table IV**). Two of the studies[15,19] evaluated disease activity through the British Isles Lupus Activity Group score. Podolskaya et al[19] reported a significant difference between scores from baseline to 1, 3, 6, and 12 months. Watson et al[15] found no significant decrease in disease activity.

### Relapses

Relapses were reported in 3 studies.[11,15,20] Overall, the relapse rate ranged from 9% to 30%, and the median time to relapse ranged from 2 to 22 months. Few studies focused on the correlation between relapses and reinstitution of B cells. Although some authors reported recurring clinical SLE relapses following repopulation of B cells,[10,17,20] Olfat et al[11] found no significant correlation between the time to CD20 B-cell reconstitution and time of relapse.

### Articular Involvement

Only 2 studies reported articular outcomes.[12,20] In 1 of these studies, 3 patients responded positively to RTX treatment for refractory polyarthritis.[12] In the other, Willems et al[20] found no improvement in arthralgia after RTX treatment.

### Renal Involvement

A total of 89 patients (33%) had nephritis before receiving RTX therapy. Significant improvement in renal function and

**Table IV. Effectiveness data**

| Studies | Follow-up | Corticosteroid dose | C3, mg/L | C4, mg/L | Hemoglobin, g/dL | Albumin, g/dL | dsDNA, IU/L | SLEDAI |
|---|---|---|---|---|---|---|---|---|
| Willems et al (2006)[20] | 2-9 mo | NR | I | I | NR | NR | D, n = 5; increase, n = 1; remain high, (n = 1) | - |
| Nwobi et al (2008)[17] | 1-3 mo | D ($P < .0001$) | I ($P < .01$) | NI | - | I ($P = .001$) | I ($P < .01$) | D ($P = .0004$) |
| Podolskaya et al (2008)[19] | 6 mo | NR | I ($P < .005$) | I ($P < .05$) | I ($P < .05$) | I ($P < .005$) | I ($P < .005$) | - |
| | 12 mo | | I ($P < .05$) | I ($P < .05$) | I ($P < .05$) | I ($P < .05$) | I ($P < .05$) | - |
| Jansson et al (2011)[14] | 6 mo | NR | I ($P = .006$) | NR | NR | NR | NR | NR |
| Su et al[10] (2012)* | 12 mo | D ($P < .001$) | NR | NR | NR | NR | NR | NR, score obviously reduced |
| Pavon-Sanchez and Sánchez-Sánchez (2013)[16] | 4 wk | NR | I ($P = .048$) | I ($P = .032$) | NI ($P = .53$) | NR | NR | D ($P = .03$) |
| Alexeeva et al (2013)[18]* | 6 mo | D ($P < .01$) | NR | I ($P = .04$) | NR | NR | I ($P = .04$) | D ($P = .002$) |
| | 12 mo | D ($P < .01$) | | I ($P < .01$) | | | I ($P < .01$) | D ($P < .001$) |
| Lehman et al (2014)[13] | 12 mo | D ($P < .05$) | I ($P < .001$) | NR | I ($P < .05$) | I ($P < .025$) | NR | D ($P < .005$) |
| Ale'ed et al (2014)[12] | 6 mo | D ($P = .005$) | I ($P = .003$) | I ($P = .01$) | NR | NR | NI ($P = .5$) | D ($P = .0002$) |
| Olfat et al (2015)[11] | 1 mo | NR | NR | NR | I | NR | NR | NR |
| Tambralli et al (2015)[9] | 12 mo | D ($P < .001$) | I ($P < .001$) | I ($P < .001$) | I ($P < .001$) | I ($P < .001$) | NI ($P = .128$) | NR |
| Watson et al (2015)[15] | 2.5 mo | D ($P = .01$) | I ($P < .001$) | I ($P = .001$) | I ($P = .02$) | I ($P = .026$) | I ($P < .001$) | NR |

-, lack of data; *D*, decreased; *dsDNA*, double stranded Desoxy-Nucleo-Adenosine; *I*, improved; *NI*, not improved; *NR*, not recorded.
*Abstract.

reduction in proteinuria were achieved within 2-12 months in 7 studies.[9,10,12,15,17-19] Jansson et al,[14] Nwobi et al,[17] and Willems et al[20] reported complete renal remission in 31%, 50%, and 25% of their cases, respectively. Partial renal remission was noted by Nwobi et al[17] and Willems et al[20] in 50% of their cases. Ale'ed et al[12] reported that 1 of their 7 patients who showed significantly improved renal function discontinued dialysis, and found persistent impaired renal function in only 1 patient. Podolskaya et al[19] reported discontinuation of renal replacement therapy in 2 patients and declining urine albumin/creatinine ratio in all patients. This improvement in urine albumin/creatinine ratio was maintained at 18 months and was observed after all treatment episodes. Increasing serum albumin levels were reported in 5 studies[9,13,15,17,19] (**Table IV**); however, Pavon-Sánchez and Sánchez-Sánchez[16] did not report any significant improvement in proteinuria.

### Mucocutaneous Involvement

Mucocutaneous manifestations of SLE were analyzed in 2 studies,[12,20] in 1 of which 3 patients treated for severe mucocutaneous lesions showed improvement, although the lesions did not resolve completely.[12] No improvement in photosensitivity was found in the other study, involving a French multicenter cohort.[20]

### Neuropsychiatric Involvement

Two studies in which neuropsychiatric manifestations were evaluated reported significant clinical improvement.[10,12] Su et al[10] noted significant improvement in delirium and cognitive disorders in 10 patients with lupus encephalopathy after 1 month of treatment, and Ale'ed et al[12] reported dramatic improvement in 2 patients with cerebritis.

### Hematologic Involvement

Thirty-two children received RTX for refractory autoimmune cytopenia (22 with thrombocytopenia and 10 with hemolytic anemia) with complete remission.[11,12,19,20] Olfat et al[11] reported complete remission in 96% of their patients after a first course of RTX, at a median of 48 days. Podolskaya et al[19] evaluated long-term outcomes in 21 patients and found significant increases in hemoglobin levels and platelet counts at 24 months ($P = .04$) and 18 months ($P = .034$), respectively (**Table IV**). Baseline lymphopenia improved significantly 6 months after RTX treatment, with sustained improvement seen at 24 months ($P = .022$).

### Immunologic Involvement

C3 and C4 values increased significantly in 8 and 6 studies, respectively.[9,12-19] This rate was higher in the short term (1-6 months) in 6 studies,[12,14-17,19] and a significant increase was seen at 12 months in 4 studies.[9,13,18,19] Eight studies[9,12,13,15,17-20] investigated the efficacy of RTX on anti–double-stranded (ds) DNA levels. In most studies, the mean or median decrease of titles was measured. Anti-double stranded Desoxy-Nucleo-Adenosine levels decreased significantly in 4 of these studies[15,17-19] and nonsignificantly in 2 studies[9,12] (**Table IV**). Ten cohort studies included an analysis of B-cell numbers after RTX infusion,[9-11,13-15,17-20] reporting that 85%-100% patients experienced complete B-cell depletion between 2 weeks and 6 months after RTX treatment. Recovery of depleted CD19/20 lymphocytes, mentioned in 5 reports,[10,11,14,17,19] began between 60 and 175 days.

### Steroid-Sparing Effect

RTX seemed to be effective in term of steroid-sparing effect. Overall, 7 of the 12 case series[9,10,12,13,15,17,18] reported a significant decrease in corticosteroid doses in the short to long term (1-12 months). Willems et al[20] reported that steroid therapy was maintained at low dosage (0.2 mg/kg/day) in 1 of their 11 patients and was tapered at 25%-50% of the baseline dosage in 5 patients. These results are summarized in **Table IV**.

### Safety

Ten studies reported adverse events in a total of 78 patients.[9-13,15,17-20] The most frequent adverse events were infusion reaction (n = 14) and viral infections (n = 10). Twenty-two patients experienced a serious adverse event, including severe cytopenia (n = 11), central nervous vasculitis (n = 1), and life-threatening infusion reaction (n = 2). Six patients developed severe infections that could be associated with RTX, including 2 patients with opportunistic infection.[10,11,17,20] Olfat et al[11] reported persistent hypogammaglobulinemia in several patients and progressive interstitial lung disease in 1 patient.

## Discussion

In this systematic review, we assessed the efficacy and safety of RTX in the management of pSLE. Although numerous clinical trials of biological therapy have been conducted in adults with SLE,[22] few studies have been conducted in children, and none of these were randomized controlled trials.

RTX has been shown to be safe and effective for treating the renal and hematologic manifestations of pSLE, especially in terms of disease activity, immunologic measures, and steroid-sparing effect. Few published studies have focused on arthralgia, photosensitivity, and mucocutaneous involvement of SLE in children, and those found little evidence of efficacy.

There was a wide heterogeneity in RTX regimens. Age and sex had no apparent effects on the pharmacokinetics of RTX.[6]

Relapse prevention is a major goal of treatment for SLE. Reviewing the efficacy of RTX in nonrenal adult SLE, relapses were reported in many studies, but the severity of relapses was milder, rates of relapses were fewer, and time to relapse was longer in the patients who received RTX therapy.[22] Resistance to B-cell depletion by RTX may be related to antibodies against the human-mouse chimeric protein.[23] It also may be genetically influenced by polymorphisms of the *FcRIIIa* gene, which is responsible for the Fc-$\gamma$ receptor required for RTX binding to phagocytes.[24]

In our review, 33% of patients had nephritis before receiving RTX therapy. Therapeutic options included hydroxychloroquine, corticosteroids, cyclophosphamide, mycophenolate mofetil, azathioprine, and methotrexate in both adult and children.[8,25,26] Treatment failure was reported in up to 43% of

patients in some studies.[27,28] In addition, even if remission is attained with immunosuppressive drugs, nephritis relapse can occur in 35% of responders and seems to be a determining factor in end-stage kidney disease in children.[28] RTX therapy reduced proteinuria even before immune changes became evident. This reduction seemed to be induced by glomeruli remodeling owing to RTX binding to podocyte membranes.[29]

In adult lupus nephritis, the Lupus Nephritis Assessment with Rituximab trial tested RTX as a first-line therapy and found negative results. Also in adults, the Exploratory Phase II/III SLE Evaluation of Rituximab trial, a randomized trial of RTX therapy in nonrenal lupus, reported negative results as well.[5] RTX is not a suitable treatment for moderate musculoskeletal, mucocutaneous, or general SLE, in which the disease manifestations are poorly correlated with SLE-related autoantibodies. For instance, anti-Ro (SSA) and anti-La (SSB) antibodies were associated with cutaneous lupus and photosensitivity but were not down-regulated by RTX.

Although primary endpoints were not attained in the LUNAR and EXPLORER trials, the clinical benefit of RTX in SLE has been noted in daily clinical practice and in open-label and retrospective studies.[22,30]

In this systematic review, RTX seemed to be effective in terms of a steroid-sparing effect. Most studies showed that children with pSLE tended to have more severe disease and worse health-related quality of life scores compared with adults. Compared with adults, children with SLE were at increased risk for nephritis, received more intensive immunosuppressive therapies, and had more adverse events related to glucocorticoid therapy.[8,31-35] Reviewing the efficacy of RTX for treating adult SLE suggested sufficient evidence of the effectiveness of RTX therapy in reducing steroid doses.

In our review, RTX appeared to be safe. Adverse events were noted in 21% of patients. In adults, adverse events were reported in a systematic review of 1 RCT (LUNAR trial),[5] 2 open studies, and 20 cohort studies.[22] The most frequent adverse events were infusion reactions and infections. The LUNAR trial reported no differences compared with placebo in terms of overall adverse events, infections, or serious infections, but differences in terms of infusion reactions, neutropenia, and herpesvirus infection.[5] The incidence of serious infections was estimated at 6.6-12.6/100 patient-years.[5,36,37] ■

*The authors express their sincere thanks to Asma Hammami for her English expertise and the improvements she made to the manuscript.*

Submitted for publication Oct 7, 2016; last revision received Jan 31, 2017; accepted May 1, 2017

Reprint requests: Ines Mahmoud, PhD, Charles Nicolle Hospital, Manar University, Boulevard 9 Avril, 1006 Bab Souika, Tunis, Tunisia. E-mail: nefmhd@yahoo.fr

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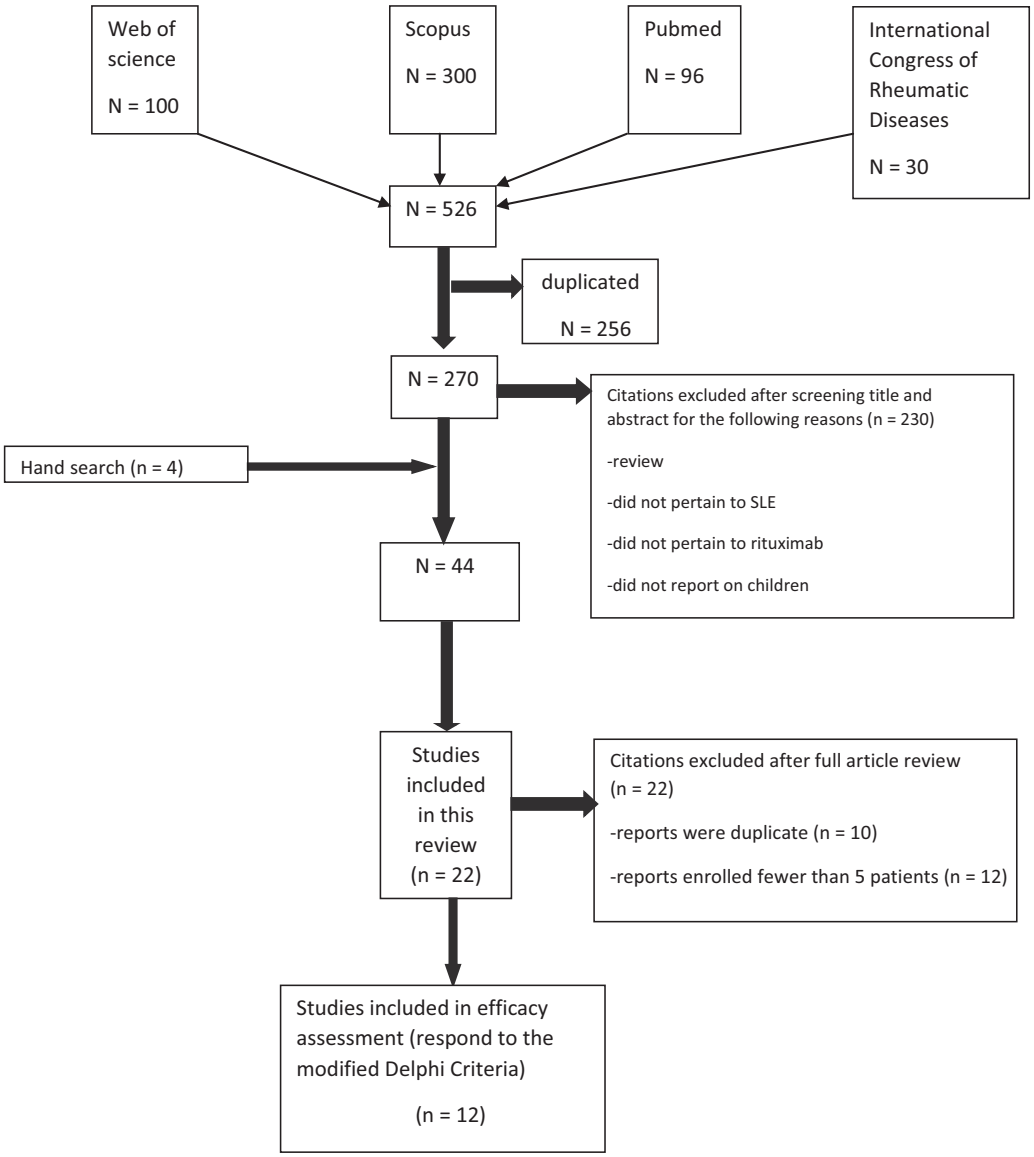


**Figure.** Flowchart showing results of the search strategy.

**Table I. Main response definitions and outcomes in the different studies**

| Studies | Main response definition | | | Follow-up/ time to flare, median (range) | Outcome | |
|---|---|---|---|---|---|---|
| | Score used | Supplemental evaluation | Time of assessment (interval) | | Proportion of CR/PR | *P* value |
| Tambralli et al (2015)[9] | Physician's global assessment | Corticosteroid use; biological parameters | 12 mo | 12 mo | | |
| Su et al (2012)[10]* | SLEDAI | Clinical impression; corticosteroid use; biological parameters | 2 wk; 1, 3, 6, and 12 mo | 12-36 mo | | |
| Olfat et al (2015)[11] | CR, defined as platelet count >100 $\times 10^9$/L (AITP) or hemoglobin ≥120 g/L (AIHA) | Corticosteroid use | 1 mo | 15-27 mo | 96% CR | |
| Ale'ed et al (2014)[12] | SLEDAI | Corticosteroid use; biological parameters | 6 mo | 6 mo | | |
| Lehman et al (2014)[13] | SLEDAI | Corticosteroid use biological parameters | 12 and 60 mo | 60 mo | | |
| Jansson et al (2011)[14] | Modified Likert scale, ranging from -5 (maximal progression of disease) to +5 (CR) | Biological parameters | 6 mo | 6 mo | 17% CR; 56% PR; 22% DP; 6% unchanged | |
| Watson et al (2015)[15] | BILAG | Corticosteroid use; biological parameters | 3 mo | 3 mo | | X |
| Pavon-Sanchez and Sánchez-Sánchez (2013)[16] | MEX-SLEDAI | Corticosteroid use; biological parameters | 4 wk | 4 wk | | X |
| Nwobi et al (2008)[17] | SLEDAI-2K | Corticosteroid use; biological parameters | 1-3 mo | 1-3 mo | CR renal: normalization of UP/Cr and eGFR<br>PR renal: >50% decline in UP/Cr | |
| Alexeeva et al (2013)[18]* | SELENA-SLEDAI | Corticosteroid use Biological parameters | 6 and 12 mo | 12 mo | | X |
| Podolskaya et al (2008)[19] | Clinical impression; BILAG | Biological parameters | 1, 6, 12, and 25 mo (BILAG); 1, 3, 6, 12, 18, 24, 30, and 36 mo (biological parameters) | 20 mo (6-38 mo) | Remission (asymptomatic patients): 52%<br>PR (some remaining symptoms after treatment): 43%<br>Some improvement (minor symptomatic benefit after treatment): 5% | |
| Willems et al (2006)[20] | CR/PR according to unknown definitions | Biological parameters | 2-9 mo | 13.2 mo (6-26 mo) | X | |

*AIHA*, autoimmune hemolytic anemia; *AITP*, autoimmune thrombocytopenia; *BILAG*, British Isles Lupus Activity Group; *CR*, complete remission; *DP*, disease unchanged; *eGFR*, estimated glomerular filtration rate; *PR*, partial remission; *SLEDAI*, Systemic Lupus Erythematosus Disease Activity Index; *UP/Cr*, urine protein to creatinine ratio.
Empty rows correspond to the lack of data and the rows with "X" signify that the study used *P* value or proportion as outcome.
*Abstract.